FUNDCHAIN

Decentralized Crowdfunding Protocol

WHITEPAPER | Version 1.0

Authored by Aditya Chotaliya
March 2026

Ethereum Sepolia Testnet • Solidity + Foundry • Next.js 14 • thirdweb v5

Live dApp	https://fund-chain-app.vercel.app
GitHub	https://github.com/adityachotaliya9299-jpg/FundChain-App
Core Contract (V2)	0xC7CF086e5ECa53BFda4D75e46753AA9ed794A131
DAO Contract (V3)	0xe5C6e9A21C1767Aa3DdC7163466C93089e354287
MultiToken (V3)	0x788d4204BAa22b4c336322F59c8E47d3a0d78C8d

Abstract

FundChain is an open, trustless crowdfunding protocol built on the Ethereum blockchain. It eliminates the need for centralized intermediaries by encoding campaign logic, fund custody, milestone governance, refund mechanics, and contributor rewards directly into auditable smart contracts. Contributors interact with campaigns via a polished Next.js 14 frontend, receive NFT reward tiers (Bronze / Silver / Gold), participate in DAO governance, and may donate using ETH, USDC, or USDT. This paper describes the protocol architecture, economic design, governance model, and technical implementation.

1. Problem Statement

1.1 Failures of Centralized Crowdfunding

Traditional crowdfunding platforms such as Kickstarter and Indiegogo are custodial by design. They hold contributor funds in escrow, charge platform fees ranging from 5–10%, and retain unilateral power to freeze campaigns or reverse payouts. Fraud is rampant: a 2023 industry report estimated that over $1 billion in crowdfunding pledges are never fulfilled annually. Backers have limited recourse and zero on-chain transparency.

1.2 Key Pain Points Addressed

- No verifiable on-chain proof that funds reached intended campaigns
- Platform can arbitrarily cancel legitimate campaigns
- Fees charged even when campaigns fail to deliver
- No automated refund mechanism — backers must rely on platform dispute resolution
- Contributors receive no governance rights or tokenized proof of participation
- Cross-border donations blocked by payment rails and KYC restrictions

2. FundChain Solution

FundChain replaces the intermediary layer with a set of composable smart contracts. Every campaign is an on-chain entity. Funds flow directly from contributor wallets to the contract. Creators cannot withdraw funds until milestones are verified by the contributor DAO. If a campaign fails or is cancelled, contributors reclaim funds autonomously via the refund module.

2.1 Core Principles

1. Trustlessness — No admin key can seize or redirect contributor funds
2. Transparency — All state transitions emit events indexed by The Graph
3. Composability — Protocol contracts are independently deployable and upgradeable
4. Incentive alignment — Creators are motivated by milestone gates; backers by NFT tiers and governance
5. Multi-asset — ETH, USDC, USDT accepted; extensible to any ERC-20

3. Protocol Architecture

3.1 Smart Contract Stack

FundChain is composed of three deployed contracts on Ethereum Sepolia:

Contract	Responsibility
FundChain.sol (V2)	Core campaign lifecycle: creation, donation, withdrawal, refund, milestones, NFT rewards
FundChainDAO.sol (V3)	Governance: proposal creation, quadratic voting, milestone approval, campaign moderation
FundChainMultiToken.sol (V3)	Multi-asset donations: ETH, USDC, USDT acceptance with oracle price normalization

3.2 FundChain.sol — Core Contract

The core contract is the campaign registry and fund custodian. It exposes the following primary functions:

Function	Description
createCampaign()	Registers a new campaign with title, description, goal, deadline, IPFS image CID, category, milestones
donate()	Accepts ETH, mints NFT reward based on donation tier, emits DonationReceived
withdrawFunds()	Transfers accumulated ETH to creator — only callable if goal met and DAO milestone approved
claimRefund()	Returns contributor's ETH if campaign failed or was cancelled
addUpdate()	Creator posts on-chain campaign update with IPFS-linked content
addComment()	Backers post comments stored on-chain with like counter
mintNFTReward()	Internal: mints Bronze (≥0.01 ETH), Silver (≥0.1 ETH), or Gold (≥0.5 ETH) ERC-721 NFT

3.3 FundChainDAO.sol — Governance Contract

The DAO contract enables decentralized milestone verification and campaign moderation. Governance token weight is proportional to total ETH donated by a contributor across all campaigns (skin-in-the-game weighting).

- Proposals: Any contributor may raise a proposal to release milestone funds
- Voting: Token-weighted yes/no voting with a 3-day window
- Quorum: 10% of total voting power must participate
- Execution: Approved proposals automatically unlock the next milestone tranche
- Moderation: DAO can flag or pause a campaign by supermajority (66%)

3.4 FundChainMultiToken.sol — Multi-Asset Donations

To maximize contributor reach, FundChain accepts ERC-20 stablecoins in addition to native ETH. The multi-token contract normalizes all donations to ETH-equivalent value using a Chainlink price feed oracle, ensuring campaign goal thresholds remain meaningful regardless of asset type.

Asset	Notes
ETH	Native asset — no approval required
USDC (ERC-20)	Circle USD Coin on Sepolia testnet
USDT (ERC-20)	Tether USD on Sepolia testnet

3.5 IPFS Storage via Pinata

Campaign images, update attachments, and metadata JSON files are stored on IPFS via Pinata's pinning service. Only the content-addressed CID is written on-chain, keeping gas costs minimal while ensuring tamper-proof storage. Files pinned to Pinata are replicated across the IPFS network and are accessible via public gateways.

4. Frontend Architecture

4.1 Technology Stack

Layer	Technology
Framework	Next.js 14 (App Router) + TypeScript
Web3 SDK	thirdweb v5 — contract reads/writes, wallet connect
Styling	Tailwind CSS + custom CSS variables
Storage	IPFS via Pinata SDK
Email	Resend API for campaign update notifications
Deployment	Vercel (automatic CI/CD from GitHub main branch)

4.2 Key Pages & Components

- Homepage (/) — Merged V1 + V2 campaign listings with status filters (Active / Funded / Ended)
- Create Campaign (/create) — 3-step wizard: details → milestones → preview & deploy
- Campaign Detail (/campaign/[id]) — Full campaign page with donation, comments, updates, milestones, DAO voting, leaderboard, charts
- Analytics (/analytics) — Real Etherscan data: funding over time, donor distribution, category breakdown with time filters (1D/5D/30D/1Y/ALL)
- Categories (/categories) — Browse campaigns by sector
- Profile (/profile) — Connected wallet's campaigns and donation history

4.3 Wallet & Transaction Flow

6. User connects wallet via thirdweb ConnectButton (supports MetaMask, WalletConnect, Coinbase Wallet)
7. Frontend reads campaign state via thirdweb useReadContract hook — no backend required
8. Donations call donate() via useSendTransaction — MetaMask signs and broadcasts
9. NFT reward minted automatically within the same transaction
10. Real-time UI updates via useContractEvents polling

5. Feature Deep-Dive

5.1 NFT Reward Tiers

Contributors who donate above threshold amounts receive ERC-721 NFT badges. These are non-transferable proof-of-contribution tokens that also double as DAO voting credentials.

Tier	Minimum Donation
🥉 **Bronze**	≥ 0.01 ETH
🥈 **Silver**	≥ 0.1 ETH
🥇 **Gold**	≥ 0.5 ETH

5.2 Milestone System

Creators define up to 5 milestones at campaign creation. Each milestone has a description and an ETH unlock amount. The DAO votes to release each tranche after reviewing creator-submitted proof (uploaded as IPFS update). This prevents creators from withdrawing the full campaign goal upfront and disappearing.

5.3 Refund Mechanism

If a campaign's deadline passes without reaching its funding goal, or if the DAO votes to cancel a fraudulent campaign, contributors can call claimRefund() at any time. The contract tracks each contributor's donation and returns the exact amount. Gas cost is borne by the contributor at time of claim — no admin action required.

5.4 DAO Governance

FundChainDAO implements a lightweight on-chain governance system purpose-built for crowdfunding oversight. Unlike general-purpose DAOs, every proposal is tied to a specific campaign action (milestone release, campaign pause, fee adjustment). This scoped governance prevents governance fatigue and keeps participation rates high.

5.5 Backer Leaderboard

Each campaign page displays a ranked leaderboard of top contributors by donation amount. This gamification layer increases average donation size by 23% in comparable Web3 crowdfunding studies. Leaderboard data is read directly from on-chain DonationReceived events — fully trustless.

5.6 Analytics Dashboard

The /analytics page fetches real transaction data from Etherscan's API and renders three chart types using Chart.js: cumulative funding over time (line), donation distribution by tier (pie), and category breakdown (bar). Time filters allow investors to view trends at different granularities.

6. Economic Design

6.1 Fee Structure

Fee Type	Rate / Destination
Platform Fee	2.5% of successfully funded campaigns — sent to DAO treasury
Failed Campaign Fee	0% — no fee charged on failed campaigns
Multi-Token Fee	0.5% additional for stablecoin donations (oracle cost coverage)
NFT Minting Gas	Paid by contributor as part of donate() transaction

6.2 DAO Treasury

Platform fees accumulate in the DAO treasury wallet. Treasury funds are governed by token-weighted DAO votes and may be used for: protocol development grants, auditor fees, gas subsidies for small campaigns, and ecosystem incentives. The treasury address is public and all disbursements are on-chain.

6.3 Incentive Alignment

- Creators: incentivized to hit milestones (funds locked until DAO approval)
- Backers: incentivized to participate (NFT rewards, leaderboard prestige, DAO governance power)
- DAO voters: incentivized to vote honestly (skin-in-the-game voting weight from donations)
- Protocol: sustainable 2.5% fee only on successes, not failures

7. Roadmap

Timeline	Milestone
Q1 2025 ☑	V1 launch — basic campaign creation & ETH donations
Q2 2025 ☑	V2 — Milestones, Updates, Refunds, NFT Rewards, Analytics
Q3 2025 ☑	V3 — DAO Governance, Multi-Token Donations, Social features
Q4 2025	Mainnet deployment — Ethereum + Polygon PoS
Q1 2026	Mobile app (React Native + thirdweb Mobile SDK)
Q2 2026	Cross-chain campaigns — Arbitrum, Base via LayerZero
Q3 2026	Fiat on-ramp integration — contributors donate with credit card
Q4 2026	Protocol governance token ($FUND) launch and DAO treasury transfer

8. Conclusion

FundChain demonstrates that trust in crowdfunding can be replaced by code. By encoding campaign rules, fund custody, milestone gates, refund logic, and governance entirely on-chain, FundChain eliminates the single points of failure that plague centralized platforms. The result is a permissionless, borderless, transparent fundraising protocol that aligns incentives for creators, backers, and the broader ecosystem.

The protocol is live on Ethereum Sepolia with all V3 features operational. The path to mainnet and cross-chain expansion is clearly defined. FundChain represents a production-grade demonstration of Web3 development capabilities across smart contract engineering, DeFi mechanics, DAO design, and full-stack decentralized application development.

Aditya Chotaliya • Web3 Developer • FundChain Protocol

https://fund-chain-app.vercel.app • March 2025